

Joe Kraemer

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EDUCATION

Georgia Institute of Technology — Atlanta, GA

Sept 2021 – Aug 2023

M.S. in Computer Science, Specialization: Machine Learning · GPA 3.90

University of Illinois at Urbana-Champaign

May 2018

B.S. in Mechanical Engineering, Minor: Spanish Language · GPA 3.61

SKILLS

Software: Python, PyTorch, Pandas, C, C#, C++, Docker, React, JavaScript, SQL, Ansible, Git, AWS, CI/CD, Spark, PLC, Protobufs, OpenThread

Coursework: Graduate Algorithms, Machine Learning, Machine Learning for Trading, AI for Robotics, Knowledge-Based AI

Languages: English (Native), Spanish (B2)

EXPERIENCE

Blue Origin — Software Engineer II

Apr 2024 – Current

- Contributed to development and maintenance of the Lunar Lander HIL (Hardware-in-the-Loop) using agile methodologies, enabling early system-level testing that integrates flight-like hardware with actual flight software for comprehensive verification before flight release.
- Developed software to emulate sensor behavior based on physics simulations, enabling flight software to interact with actual system components in real time.
- Automated and standardized deployment of configurations across multiple servers using Ansible, ensuring consistency and stability of the VTB environment across all testing runs.
- Collaborated with cross-functional teams to integrate hardware and software from diverse sources, troubleshooting complex system-interaction and hardware-specific issues.
- Integrated the VTB into customer CI/CD pipelines, enabling continuous testing and validation of flight software and early identification of issues.

DMC Inc — Systems Engineer II

Jun 2020 – Feb 2022

- Managed and executed all phases of the project life-cycle: sales, customer relationship, requirements and specifications, architecture, hardware selection, programming, testing, onsite deployment, and post-project support.
- Enhanced radio communication and client request handling with parallelization (mbed thread API) and a priority queue for critical data packets, lowering latency and raising throughput.
- Led CI/CD pipeline development including static analysis, unit testing, and on-metal tests using mbed's Icetea framework to evaluate real-world radio performance and publish results to the commit as residuals.
- Developed firmware for a PIC32 PID motor controller: state machine, drivers, and modules for pressure sensors, EEPROM (I2C), UART, and PWM-controlled H-bridge.

DMC Inc — Systems Engineer I

Sep 2018 – Jun 2020

- Developed large-scale factory automation solutions across many industries using Siemens and Allen-Bradley PLCs and a variety of system platforms.

GRADUATE COURSEWORK HIGHLIGHTS

- **Deep Learning** — CNNs, RNNs, transformers; final project identified identical e-commerce product listings via text + image embeddings, clustered by similarity, trained on AWS SageMaker GPU instances.
- **Reinforcement Learning** — Deep Q-Learning agent for Lunar Lander (Experience Replay, Target Networks); PPO and QMIX for multi-agent RL in Google Football; hundreds of parallel hyperparameter trials with Ray Tune.
- **Data & Visual Analytics** — big-data collection and visualization with R, D3, Spark, Hadoop, OpenRefine; PageRank and Random Forest; Spark/Scala on AWS EMR, Databricks, Azure, GCP.

PERSONAL PROJECTS

- **AWS Stock Video Tagger** — asynchronously uploads videos to S3 and starts Rekognition jobs, saving highest-confidence tags to CSV for upload to stock-video sites; parallelized with threading and AWS SNS.

INTERESTS

Photography, Climbing, Sim Racing, Hiking, Snowboarding, Board Games, Running, Traveling